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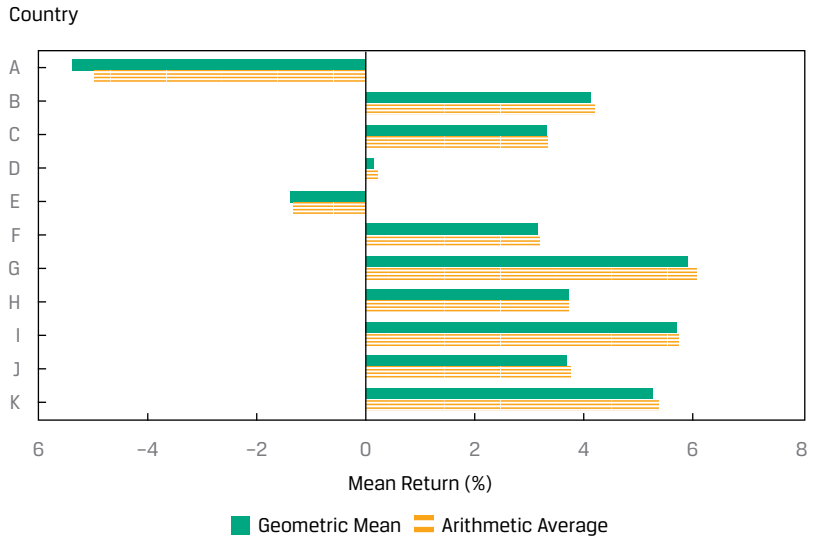
Exhibit 6: Arithmetic and Geometric Mean Returns for Country Stock Indexes, Years 1 to 3



The geometric mean return represents the growth rate or compound rate of return on an investment. One unit of currency invested in a fund tracking the Country B index at the beginning of Year 1 would have grown to (1.078)(1.063)(0.985) = 1.128725 units of currency, which is equal to 1 plus Country B's geometric mean return of 4.1189 percent compounded over three periods: $[1 + 0.041189]^3 = 1.128725$. This math confirms that the geometric mean is the compound rate of return. With its focus on the actual return of an investment over a multiple-period horizon, the geometric mean is of key interest to investors. The arithmetic mean return, focusing on average single-period performance, is also of interest. Both arithmetic and geometric means have a role to play in investment management, and both are often reported for return series.

For reporting historical returns, the geometric mean has considerable appeal because it is the rate of growth or return we would have to earn each year to match the actual, cumulative investment performance. Suppose we purchased a stock for EUR100 and two years later it was worth EUR100, with an intervening year at EUR200. The geometric mean of 0 percent is clearly the compound rate of growth during the two years, which we can confirm by compounding the returns: $[(1 + 1.00)(1 - 0.50)]^{1/2} - 1 = 0\%$. Specifically, the ending amount is the beginning amount times $(1 + R_G)^2$.

However, the arithmetic mean, which is [100% + −50%]/2 = 25% in the previous example, can distort our assessment of historical performance. As we noted, the arithmetic mean is always greater than or equal to the geometric mean. If we want to estimate the average return over a one-period horizon, we should use the arithmetic mean because the arithmetic mean is the average of one-period returns. If we want to estimate the average returns over more than one period, however, we should use the geometric mean of returns because the geometric mean captures how the total returns are linked over time.

The Harmonic Mean

The **harmonic mean**, $\bar{X}_H$, is another measure of central tendency. The harmonic mean is appropriate in cases in which the variable is a rate or a ratio. The terminology "harmonic" arises from its use of a type of series involving reciprocals known as a harmonic series.